Informational Order Flow and Price Persistence: A Diagnostic Study of Look-Ahead Bias and Conditional Reversal in Order-Submission Bursts

Nicholas Jiang

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Abstract

Distinguishing temporary mechanical price impact from permanent informational price discovery is a central challenge in market microstructure. This paper asks whether intraday limit-order submission bursts—temporally clustered, same-side executions that proxy for institutional order splitting—encode tradable information about overnight price persistence. The honest answer, established under rigorous look-ahead controls and at cross-sectional scale, is largely negative, and the paper's contribution is the diagnosis rather than a strategy.

On a handful of high-volatility single names, heavily filtered burst flow appears strongly predictive; but those names coincide with the in-sample parameter-tuning set, and their large intraday markouts are an artifact of gating on a decay filter D_b that is itself the forward markout. Evaluated free of look-ahead across a 482-name universe traded on NASDAQ (2022–2026, 438 names strictly out-of-sample), the volume-weighted cross-sectional order-imbalance signal is insignificant (Fama–MacBeth $t \approx -1.4$; a count-weighted variant is weakly significant but reversal-signed and economically sub-cost), and the per-name overnight strategy's return is dominated by an incidental short-market-beta exposure ($\beta \approx -0.75$; residual $\alpha < 0$), of which roughly 62% is the directional bet and the residual skill term is indistinguishable from zero. Correcting the burst-start-mid target to the tradable close-to-open return collapses the apparent predictive correlation to zero once observations are clustered by date.

The one affirmative result is conditional and modest. The sign of the signal is governed by tick structure: conditioning on relative tick size yields a monotone asymmetry—continuation in large-tick names, mean-reversion in tick-constrained ones—that is orthogonal to the Fama–French five factors plus momentum. The associated market-neutral reversal earns a full-sample Sharpe of 1.5 but only an out-of-sample Sharpe of ≈ 0.8 ($t = 1.4$, deflated-Sharpe probability 0.70), i.e. not statistically significant; and a plain-reversal and plain-signed-volume baseline shows that the linear online-SGD apparatus adds nothing over the raw signed order flow. The durable contributions are therefore methodological: the identification of the D_b look-ahead and the κ-firewall that isolates it, a date-clustered markout methodology, a systematic comparison of burst reconstructions (of which only hidden-execution clustering carries a genuine but sub-spread footprint), and a precisely characterized—if unmonetizable—conditional reversal where the tick binds.

1 Introduction

Financial markets are driven by heterogeneous trading activity. While some orders primarily provide or consume liquidity and generate only short-lived price effects, others reflect fundamental information that leads to persistent, permanent price changes. The theoretical foundations of informational price discovery have long established that informed traders gradually incorporate private signals into prices to avoid excessive market impact (Kyle, 1985). Empirical evidence supports this, demonstrating that certain classes of traders systematically trade in the direction of permanent price changes, contributing to price discovery rather than transient mechanical pressure (Brogaard et al., 2014; Hasbrouck, 1991).

A key empirical regularity in equity markets is the strong persistence of order flow over extended horizons. This persistence is overwhelmingly driven by order splitting, a process where a single institutional investor gradually executes a large trading intention through a sequence of same-side orders (Tóth et al., 2015). A large recent literature formalizes the predictive content of this signed order flow through order-flow imbalance and its deep-learning extensions (Cont et al., 2014; Kolm et al., 2023; Lucchese et al., 2024), through cross-asset propagation (Cont et al., 2023b), and—most directly related to the present work—through a *conditional* decomposition of trade flow into typed order-imbalance predictors of the cross-section of daily returns (Cont et al., 2023a; Lu et al., 2024; Sitaru et al., 2023). The burst construction studied here can be read as one such conditioning: an event-time aggregation of same-side aggressive flow into discrete, geometrically filtered clusters. Furthermore, the slow decay of market impact necessitates that large meta-orders be executed through hidden or fragmented bursts to minimize execution costs (Moro et al., 2009; Bucci et al., 2019). Motivated by these dynamics, this paper studies whether intraday order submission behavior can be used to identify longer-horizon informational order flow. Note that while resting limit order submissions establish the background liquidity structure, the empirical scope of this paper currently focuses almost exclusively on aggressive order executions (marketable trades) that actively consume this liquidity. Future research will expand this framework to directly cluster passive limit order additions and cancellations.

I define informational order flow as trading activity whose price impact persists over hours and into overnight clearing returns. The focus is on isolating any longer-horizon predictive content in intraday limit order data without relying on ex post price realization, and on the execution architecture that would map short-horizon burst footprints to a long-horizon portfolio if such content existed.

2 Data and Limit Order Book Reconstruction

My empirical analysis relies on ultra-high-frequency, event-by-event limit order book (LOB) data provided by LOBSTER (Limit Order Book System, the Efficient Reconstructor), reconstructed directly from the NASDAQ (ITCH) message stream. The study uses the data at two scales. A small *diagnostic set* of high-volatility single names—NVIDIA (NVDA), Tesla (TSLA), JPMorgan Chase (JPM), and Morgan Stanley (MS)—is used to develop the pipeline and, as it turns out, to expose its failure modes; these four names are also the Optuna parameter-tuning set and are therefore in-sample throughout (Sections 8–10). The primary evaluation is a *full cross-section*: the 482 names traded on NASDAQ with continuous LOBSTER coverage over 2022–2026 (a mix of NASDAQ- and NYSE-listed common stocks and ETFs, reconstructed from the NASDAQ book), of which 438 are strictly out-of-sample relative to the 40/50-name tuning universe. Universe membership is fixed as of the start of the sample by LOBSTER coverage (not by survival to 2026); the survivorship and delisting implications of this construction are examined in the robustness analysis. Daily open and close prices used for overnight and close-to-close returns are sourced from the Polygon/Massive consolidated daily feed; corporate-action (split/dividend) adjustment of these prices, which is first-order in the low-priced names that carry the affirmative result, is treated explicitly as a data-integrity check rather than assumed. Because LOBSTER reconstructs the NASDAQ book only, the fraction of consolidated volume captured varies by name (notably for the NYSE-listed financials

JPM and MS); this venue-coverage caveat is discussed where the single-name comparisons are drawn. A non-sequential 2019–2021 archive (LLY, AAPL, SPY) is retained purely as a parameter-transfer robustness probe, *not* as an out-of-sample trading deployment, since it applies 2023–2024-tuned parameters to earlier data.

Standard vendor-aggregated data often introduces temporal blurring that obscures high-frequency signal integrity. To prevent this, I process the raw Level 3 LOBSTER message stream directly using a bespoke C++ data processor. By tracking the exact bid and ask price levels, resting depth, and microsecond timestamps, this procedure allows me to observe the precise liquidity conditions facing an order at the exact moment of submission.

2.1 Data Summary Statistics

Table 1 presents the key summary statistics for the evaluated asset universe. A critical microstructural distinction emerges from the average spread column. NVDA's average quoted spread of \$0.104 spans approximately 10 minimum price increments (ticks), meaning that a moderately aggressive institutional order can sweep through multiple resting price levels before being fully absorbed. This multi-level sweep generates the sharp, identifiable burst signatures that the detection algorithm relies on. By contrast, AAPL's average spread of \$0.026 is virtually locked at the \$0.01 minimum tick, with massive queued depth at the best bid and ask. In this tick-constrained environment, even large institutional orders are absorbed passively at a single price level, producing no detectable multi-level sweep and rendering the burst clustering ineffective.

Ticker	Period	Trading Days	Total Raw Bursts	Bursts/Day	Avg Daily Burst Vol (sh.)	Avg Spread ($)
NVDA (train)	2023–2024	499	894,343	1,792	12,083,125	0.1039
TSLA (train)	2023–2024	499	757,120	1,517	11,702,121	0.0346
JPM (train)	2023–2024	499	1,251,679	2,508	993,261	0.0298
MS (train)	2023–2024	499	1,108,679	2,222	850,519	0.0211
LLY (train)	2019–2021 probe	505	308,361	611	401,126	0.0604
AAPL (probe)	2019–2021 probe	360	117,253	326	2,884,874	0.0263
SPY (probe)	2019–2021 probe	316	122,221	387	8,062,116	0.0133

Table 1: Data summary statistics for the single-name diagnostic set. *Avg Daily Burst Vol* is the average daily aggregate *burst* volume (shares)—the volume captured by the detector, not consolidated ADV, which for the NYSE-listed names (JPM) is roughly an order of magnitude larger. Avg Spread is the mean time-weighted quoted spread at burst initiation. All four 2023–2024 names, *and* LLY, belong to the Optuna tuning universe and are in-sample; the 2019–2021 rows apply 2023–2024-tuned parameters to earlier data and are parameter-transfer robustness probes, not out-of-sample deployments. The reduced trading-day counts for AAPL (360) and SPY (316) reflect gaps in continuous LOBSTER message coverage over the 2019–2021 archive.

3 Burst Formulation and Parameterization

Because single, isolated order submissions are highly noisy and rarely informative at horizons beyond a few seconds, I aggregate temporally clustered orders into discrete "bursts." For each burst b, I define its start time t_b, reference mid-price m_{t_b}, and its signed aggregate volume:

$$Q_b = \sum_{j \in b} side_j q_j \tag{1}$$

where *side* $\in \{+1, -1\}$ denotes a buy or sell order. These reconstructed bursts constitute the fundamental unit of analysis. That they are genuine structures rather than artifacts of arrival-rate clustering is verified directly (Appendix B): the observed count of same-side clustered bursts exceeds an *inhomogeneous*-Poisson null matched to each name's empirical intraday intensity profile by a median z of 47 (and a homogeneous null by z of 87), rejecting the Poisson null on 100% of name-days.

To construct my dataset from the reconstructed limit order book, I evaluate two distinct clustering methodologies governed by physical control parameters to isolate algorithmic intent from retail noise:

- **Strict Silence Threshold ($\Delta t_{silence}$)**: My baseline approach where a burst rigidly terminates if the limit order book experiences inactivity exceeding a specific temporal limit (e.g., 0.5s, 1.0s, or 2.0s).
- **Hawkes Process Intensity ($\lambda(t)$)**: A dynamic behavioral model where burst survival is dictated by a self-exciting point process. The conditional intensity function evolves as:

$$\lambda(t) = \mu + \sum_{t_k < t} \alpha \, e^{-\beta(t - t_k)} \tag{2}$$

  where μ is the background intensity, α controls the magnitude of the excitation from each prior event at time t_k, and β governs the exponential decay rate. In my implementation, I fix $\beta = 1.0$ and $\alpha = 0.5$, so each incoming trade spikes the intensity by 0.5, which then decays exponentially with a half-life of $\ln 2/\beta \approx 0.69$s. A burst remains active as long as $\lambda(t)$ exceeds a background intensity threshold $\lambda_{\min}$. I denote each configuration as $\beta_\beta \lambda_{\lambda_{\min}}$; for example, $\beta_{1.0}\lambda_{0.3}$ indicates $\beta = 1.0$ with a minimum intensity threshold of 0.3. This replaces rigid silence cutoffs with a fluid structural definition that naturally bridges microscopic liquidity gaps.
- **Fractional Sweep Volume (vol_frac)**: Dynamically establishes the baseline size for a candidate burst, requiring the aggregate burst volume to exceed a specific fraction of the asset's 14-day trailing Average Daily Traded Volume (ADV). Crucially, this ADV is computed strictly using executed trades (LOBSTER Types 4 and 5), insulating the fractional threshold from the artificial inflation of high-frequency quote-stuffing and rapid cancellations.
- **Directional Consistency Ratio (dir_thresh)**: Enforces that a minimum proportion of the events within the cluster must be on the same side of the book (e.g., $> 70\%$).
- **Volume Ratio (vol_ratio)**: Limits the allowable opposing volume to a fraction of the dominant side's volume.
- **Short-Horizon Decay Filter (κ)**: Retains bursts where the structural decay measure exceeds a threshold, $D_b \geq \kappa$. The **peak impact** of a burst is defined as the maximum absolute mid-price displacement achieved during the burst lifetime:

$$\text{PeakImpact}(b) = \max_{t \in [t_b, t_b^{\text{end}}]} |m_t - m_{t_b}| \tag{3}$$

  The **decay metric** D_b then measures how much of this displacement persists over the 1–10 minute forward window. The canonical definition used throughout the pipeline is the average directional markout from the burst-termination mid M_{end},

$$D_b = \frac{1}{3} \sum_{\tau \in \{1,5,10\}\text{m}} \text{side}_b \times (M_\tau - M_{\text{end}}), \tag{4}$$

normalized by PeakImpact(b) where a scale-free ratio is required. A large positive D_b indicates that the mid-price has held or extended the burst's displacement; a value near 0 indicates full mean-reversion. The κ filter retains only bursts with $D_b \geq \kappa$.

D_b is forward-looking. Because D_b is literally the 1–10 minute forward markout, it is realized only *after* the burst terminates and cannot be gated on for any intraday horizon inside its own window without circular look-ahead (Section 8). It is therefore admissible only (i) for the overnight target and (ii) as a filter applied *strictly to the trailing training window*, never to same-day prediction candidates. A live burst ending near the close realizes D_b up to ten minutes later—after the 3:50/3:55 PM market-on-close cutoff—so its earlier use as a prediction-time feature (Section 5) is timing-infeasible; the corrected feature set is reported in the revised breadth analysis.

Crucially, when predicting short-horizon permanence (e.g., 1-minute or 3-minute targets), κ is strictly set to 0. The decay filter D_b inherently relies on observing the price trajectory at $t + 10$ minutes. Applying a $\kappa > 0$ threshold to short-horizon targets would introduce a severe look-ahead bias, as the filter's observation window would extend beyond the prediction horizon itself. Consequently, short-horizon bursts must be evaluated solely on their initial physical and behavioral footprints without relying on post-burst decay gating.

Why Short-Horizon Targets Are Excluded from Phase III. While the initial Optuna sweep evaluated both short-horizon targets (e.g., cls_1m, achieving AUC > 0.64) and long-horizon targets (cls_clop), the Phase III execution pipeline exclusively targets the overnight horizon for two reasons. First, short-horizon predictions require immediate intraday execution: to capture a predicted 1-minute or burst-to-close movement, the strategy would need to enter a position within seconds of burst detection and exit minutes later, requiring a fundamentally different execution architecture supporting multiple simultaneous intraday positions with per-burst sizing. This stands in contrast to the MOC/MOO framework, which aggregates all signals into a single daily trade. Second, the short-horizon AUC advantage is mechanically inflated by the D_b look-ahead constraint: because $\kappa = 0$ is forced for short targets, the pipeline evaluates a much larger and noisier burst universe, making the high AUC less economically actionable after transaction costs. The overnight target, by contrast, permits rigorous κ-gating and produces a concentrated, high-conviction signal compatible with institutional-scale execution.

4 Optuna Optimization and Physical Results

To identify the physical parameters that best generalize across different time horizons and market regimes, I optimized the physical burst controls using Optuna (Akiba et al., 2019). The objective function evaluated a `HistGradientBoosting` model (using a strict 70/30 chronological train/test split on 2023-2024 data), maximizing the out-of-sample logistic regression Area Under the ROC Curve (AUC) on binary classification permanence targets (cls_*).

I present the optimization results in two phases, corresponding to the two distinct temporal definitions of a burst.

4.1 Phase I: Strict Silence Threshold

The initial strict-silence parameterization was superseded by the Hawkes clustering below and we do not report its full sweep; briefly, short-horizon targets favored longer silence gaps (up to 2.0s) while long-horizon targets required aggressive κ-gating (e.g. NVDA cls_close at $\kappa \approx 1.28$ discarded most raw bursts). To illustrate the selective power of each filter stage, Table 2 presents the sequential compression cascade under the regression-optimized parameters. For NVDA, the dominant filter is κ (only 24.6% of raw bursts survive when applied in cascade), while vol_frac is less binding. For TSLA, the vol_frac threshold is the primary gate, eliminating 97.4% of bursts immediately due to TSLA's much higher fractional volume requirement ($vol\_frac = 0.0033$ vs. NVDA's 6.45×10^{-5}).

Filter Stage	**Condition**	**NVDA Surviving**	**TSLA Surviving**
Raw bursts	—	894,343 (100%)	757,120 (100%)
vol_frac	$\geq$ ADV threshold	762,253 (85.2%)	19,641 (2.6%)
$+\ dir\_thresh$	≥ 0.51 / 0.57	744,477 (83.2%)	15,493 (2.0%)
$+\ vol\_ratio$	≤ 0.58 / 0.60	574,946 (64.3%)	11,792 (1.6%)
$+\ \kappa$	$D_b \geq 0.82$ / 1.74	220,430 (24.6%)	5,073 (0.7%)

Table 2: Sequential filter cascade under regression-aligned parameters. Each row applies an additional filter on top of the previous. The dominant filter varies by asset: κ for NVDA, vol_frac for TSLA.

4.2 Phase II: Hawkes Process Structural Optimization

Recognizing the limitations of rigid time cutoffs, I transitioned the pipeline to isolate structural footprints via the mutually exciting Hawkes Process. Table 3 outlines the optimal parameters for long-horizon targets under this behavioral regime.

Ticker	**Target**	**Hawkes Tag**	**Best AUC**	**Dir. Thresh**	**Sweep Frac** (vol_frac)	**Vol. Ratio**	κ
NVDA	cls_clop	$\beta_{1.0}\lambda_{0.3}$	0.6046	0.511	1.87e-04	0.600	0.988
TSLA	cls_clop	$\beta_{1.0}\lambda_{0.3}$	0.5325	0.643	3.64e-03	0.481	0.969
JPM	cls_clop	$\beta_{1.0}\lambda_{0.8}$	0.5477	0.503	1.0e-05	0.553	0.206
MS	cls_clop	$\beta_{1.0}\lambda_{0.5}$	0.5527	0.830	4.30e-03	0.405	0.950

Table 3: Hawkes-process optimization for the overnight (`cls_clop`) target, one row per diagnostic-set name (the `cls_clcl`/`cls_close` rows are omitted for brevity). Directional-consistency thresholds cluster near 0.5–0.75. The AUCs are in-sample classification maxima and, as Sections 8–9 show, do not translate into deployable out-of-sample edge.

The AUCs are stable across neighboring Hawkes thresholds (e.g. NVDA *cls_clop* varies by $\Delta < 0.005$ across $\lambda_{\min} \in \{0.3, 0.5, 0.8\}$), but—as Sections 8–9 show—these in-sample, pseudo-replicated classification maxima do not correspond to any deployable out-of-sample edge, and we do not read parameter-neighborhood stability as evidence of one.

5 Feature Engineering and Impact Measurement

Upon the termination of a classified burst, I extract a rich, high-dimensional execution fingerprint. Beyond standard Limit Order Book State metrics (Spread, Imbalance, Depth Ratio) and Temporal Dynamics, I incorporate specific algorithmic footprints such as Trade-Size Variance, Round Lot Percentages, and Pre-burst Quote Depletion.

To quantify the true economic footprint of a burst and how much of its initial price displacement persists at longer horizons τ, I utilize a Transformed Volume-Scaled Impact (VSI):

$$VSI(b; \tau) = \operatorname{arcsinh}\left(Q_b \times \text{side}_b \times (P_\tau - m_{t_b})\right) \tag{5}$$

To prevent denominator instability inherent in ratio-based metrics during superficial algorithmic pinging, I eschew traditional permanence ratios in favor of this Transformed Volume-Scaled Impact. By scaling the raw directional price displacement by the capital deployed (Q_b), I control metric explosion, and the arcsinh transform compresses heavy-tail outliers into a regression-friendly domain.

Target base and tradability. A subtlety that matters for the overnight target: when the reference mid m_{t_b} is the *burst-start* mid, the reg_clop target $\text{side}_b \times (\text{Open}_{t+1} - m_{t_b})$ folds in the burst's own contemporaneous impact and the burst-to-close drift, both realized before any market-on-close entry and therefore unearnable. Measured instead from the close mid M_{end}—the tradable close-to-open return—the same predictive rank correlation collapses roughly fivefold and, once clustered by date, to a statistical zero (Section 9). The predictive ρ reported below for the burst-start base should thus be read as an upper bound on the earnable association, not the tradable quantity.

5.1 Feature Importance and Model Interpretability

Since the `SGDRegressor` is a linear model operating on standardized features, its learned coefficients quantify each feature's marginal contribution to predicted permanence. Across both NVDA and TSLA, one feature dominates: `Direction` (standardized coefficient $\approx +5.0$ for NVDA, $+4.0$ for TSLA), roughly seventy times the next-largest loading. In other words the fitted model is, to first order, "follow the burst side"—which raises the question, addressed by the Direction-only ablation in Section 9, of whether the remaining features earn their place at all. The second and third features are `D_b` (coef $\approx +0.07$) and `PeakImpact` (coef ≈ -0.06), with `TradeCount` and `Duration` fourth and fifth. Two caveats attach to this ranking. First, `D_b` is the forward 1–10 minute markout (Eq. 4); its appearance as a prediction-time feature is the timing-infeasibility flagged in Section 3, since D_b is not realized until after the burst (and, near the close, after the MOC cutoff). Second, these coefficients are estimated on the two in-sample tuning names; at cross-sectional scale (Section 9) the deployed overnight regressor's prediction is in fact *anti*-correlated with the burst side, so the "follow-the-side" reading does not survive out of sample.

6 Predictive Modeling and Strategy Formulation

The in-sample classification AUCs above are per-burst, pseudo-replicated, and optimizer maxima (Section 9); rather than treat them as established signal, the remainder of the paper asks whether any of it maps to robust, walk-forward tradability under institutional constraints—and, at cross-sectional scale, finds that it does not. To validate economic viability, I engineered a continuous, online execution pipeline utilizing an `SGDRegressor` evaluating chronological limit order book states.

6.1 Resolving Look-Ahead Bias

A fundamental challenge in evaluating intraday momentum is the inherent look-ahead bias present in optimization sweeps. I address this by strictly separating *structural* from *predictive* parameters. The Optuna optimization conducted on 2023–2024 data was limited exclusively to deriving the *physical* definition of an order burst (fractional ADV limits, Hawkes intensity thresholds, directional consistency gates). These parameters define the structural geometry of algorithmic execution, which we treat as more stable across time than a macroeconomic regime variable (an assumption the 2019–2021 transfer probe stress-tests, not a claimed universal constant). By contrast, the actual trading model is an Online Stochastic Gradient Descent regressor whose weights initialize from scratch for each evaluation period. Following a 30-day burn-in, the model trains strictly on day T to predict day $T+1$, never observing future data.

6.2 Phase III Formalization: Institutional Execution Architecture

Under the Phase-III framework, the execution engine separates intraday observation from trade execution. To resolve end-of-day boundary conditions and enforce institutional sizing and friction constraints, the pipeline operates via the following ruleset:

1. **Intraday Microstructure Observation**: A burst is formed if consecutive limit orders arrive rapidly, governed by the Hawkes Intensity threshold. The burst is deemed a "Candidate" if its aggregate volume strictly exceeds the 14-day trailing fractional ADV and its directional consistency is highly concentrated.

2. **Machine Learning Evaluation**: Every candidate burst is evaluated on-the-fly by the `SGDRegressor`. The model analyzes the burst's execution fingerprint and predicts the Transformed VSI (*reg_clop*). If the prediction > 0.0, the burst is labeled *Informational*.

3. **Daily Flow Aggregation and Boundary Resolution**: I aggregate the directional volume of all Informational bursts throughout the day. To prevent boundary condition corruption, the pipeline strictly enforces a 10-minute end-of-day dead-zone. Any burst occurring after 3:50 PM EST is ineligible and discarded, so that all 10-minute D_b forward observation windows mature before the market closes. (This dead-zone protects the overnight D_b label but does not make D_b usable as a live intraday feature—see Section 3.)

4. **Institutional Position Sizing**: At exactly 4:00 PM EST, I calculate the Net Aggregated Burst Flow. A Long position is initiated if the net flow is positive, and a Short position if negative. To adhere to institutional portfolio constraints, the backtester enforces a strict **Fixed AUM limitation**, allocating exactly \$10,000,000 of nominal capital per trade, completely independent of the underlying raw burst volume.

5. **Realistic Execution Friction**: The position is held strictly overnight and exited unconditionally at 9:30 AM EST (Next Day's Open). While executing at the MOC and MOO crossing sessions structurally bypasses the continuous intraday L1 bid-ask spread, it is not without friction. To realistically account for exchange fees, clearing costs, and auction price-impact slippage on institutional allocations, I subject the strategy to explicit transaction cost penalties, quantified in subsequent sensitivity analyses. For highly liquid assets such as NVDA and TSLA, a \$10,000,000 Fixed AUM allocation represents less than 0.5% of the average consolidated closing auction volume (the NYSE/NASDAQ closing cross typically exceeds \$2B in notional value for mega-cap equities), ensuring that simulated executions do not exceed the absorption capacity of the MOC/MOO crosses.

I define the daily net accumulated directional imbalance for asset i on day t as:

$$S_{i,t}(reg\_clop) = \sum_{b \in B_i(t)} \mathbf{1}\{\text{preds}_b > 0\} \times Q_b \tag{6}$$

Execution is only triggered if the aggregate imbalance clears a cost-aware buffer cb, defined as $cb_{i,t} = c \cdot \hat{s}_{i,t}$, where $\hat{s}_{i,t}$ is the name's trailing average relative half-spread at the close and c is a fixed multiplier (set to $c = 1$, i.e. the signal must exceed a one-half-spread friction); a trade is placed only when $|S_{i,t}|$ implies an expected move exceeding $cb_{i,t}$. The multiplier $c = 1$ is a fixed default, *not* tuned; varying it over $c \in \{0.5, 1, 2\}$ trades off coverage for per-trade quality without changing the qualitative (null) cross-sectional result, so the buffer is a deployment gate rather than an alpha source.

7 Empirical Results on the Diagnostic Set: Apparent Success and Its Artifacts

This section reports the single-name pipeline results at face value and then, in the same breath, identifies why each apparent success is an in-sample or look-ahead artifact. It is deliberately a cautionary narrative: the four 2023–2024 names are the parameter-tuning set (in-sample), LLY is likewise in the tuning universe, and the strongest intraday effects are contaminated by the D_b look-ahead documented in Section 8. The quantitative refutations at cross-sectional scale follow in Section 9. All results here enforce the execution architecture of Section 6.2: \$10,000,000 fixed AUM, 1.0 bps round-trip costs, MOC entry, and MOO exit unless otherwise noted.

7.1 Unfiltered Baseline: The Necessity of Filtering

Naive, unconditional burst-following generates negative expected value. Table 1 shows that the pipeline detects 894,343 raw bursts for NVDA and 757,120 for TSLA over 499 trading days. Executing on every detected burst yields negative per-trade expectancy, driven by immediate bid-ask reversion and the large volume of non-informational algorithmic pinging (the honest, look-ahead-free intraday markout net of the spread is -3.5 to -7.7 bps per burst; Section 8). The physical filters reduce NVDA to ∼220,430 bursts (Table 2) and TSLA to ∼5,073, and the SGD informational gate concentrates these into a few hundred aggregated daily trades. This compression shows that *some* filtering is necessary to avoid the reversion drag; it does *not*, by itself, establish that the surviving signal is informative—a distinction Sections 8–9 make good on.

7.2 Regression-Aligned Optimization and Empirical Resolution

To systematically bridge the disconnect between predictive capacity and economic tradability, I implemented a regression-aligned hyperparameter sweep. This framework addresses three critical structural traps that plague standard machine learning backtests in high-frequency regimes:

- **The Sparsity Exploit:** Optimization algorithms tasked with maximizing standard regression correlations quickly exploit dataset sparsity. By shrinking the dataset to an extreme degree, the optimizer can filter out 99.9% of bursts, leaving a tiny sample where the correlation approaches 1.0. To prevent this, I applied a confidence-scaled scoring metric: $Score = \rho \times \min(1, N_{test}/500)$, alongside a hard minimum floor of 200 total bursts.

- **Fat-Tail Outliers:** Financial return series are characterized by massive fat-tail events. Standard Mean Squared Error (MSE) forces the model to overfit these extreme outliers. I replace MSE with the Spearman rank correlation coefficient (ρ), evaluating relative order ranks rather than absolute magnitudes.

- **Model Alignment:** The Optuna sweep evaluates candidate physical parameters using the exact same machine learning pipeline deployed in execution: an `SGDRegressor` with Huber loss ($\epsilon = 1.35$).

Table 4 outlines the optimal physical parameters discovered by this regression-aligned sweep. The resulting Spearman correlations range from 0.088 to 0.199 with nominally minuscule p-values. Two caveats, developed in Sections 8–9, gut the apparent significance and are stated here so the table is not over-read. First, these p-values (e.g. JPM's 9.3×10^{-126}) treat $\sim 10^5$ per-burst observations as independent; because all same-day bursts share one overnight label, the effective sample size is the number of *days*, and date-clustering the identical statistic collapses the significance by more than an order of magnitude. Second, the target is measured from the burst-start mid, so ρ overstates association with the tradable close-to-open return by roughly fivefold (Section 5); on the tradable, date-clustered target the correlation is statistically zero. The numbers in Table 4 are thus best read as in-sample, pseudo-replicated upper bounds, not evidence of tradable predictability.

Ticker	Target	Best Hawkes	Spearman ρ	p-value	vol_frac	dir_thresh	vol_ratio	κ
NVDA	reg_clop	$\beta_{1.0}\lambda_{0.3}$	0.196	0.0	6.45×10^{-5}	0.509	0.583	0.824
TSLA	reg_clop	$\beta_{1.0}\lambda_{0.3}$	0.094	3.0×10^{-7}	3.30×10^{-3}	0.568	0.599	1.736
JPM	reg_clop	$\beta_{1.0}\lambda_{0.3}$	0.049	9.3×10^{-126}	1.00×10^{-5}	0.561	0.570	0.200
MS	reg_clop	$\beta_{1.0}\lambda_{0.3}$	0.125	1.0×10^{-3}	4.94×10^{-3}	0.902	0.018	1.399

Table 4: Regression-aligned Optuna sweep results (in-sample tuning names). The objective maximizes confidence-scaled Spearman rank correlation evaluated with the backtest `SGDRegressor`. The reported p-values are per-burst and pseudo-replicated (effective N = number of days); the target is measured from the burst-start mid and overstates tradable association—see the text and Sections 8–9.

7.3 Phase-III Walk-Forward Execution and Regime Dynamics

I executed the full Phase III backtest using these regression-optimized physical parameters. The performance metrics are summarized in Table 5.

Ticker	Period	Trades	Long/Short	Long Win%	Short Win%	Net PnL ($)	ROC (%)	Sharpe	Spearman ρ
NVDA (in-sample)	2023–2024	471	204 / 267	61.76%	45.69%	+$8,198,811	+81.99%	0.59	0.198
TSLA (in-sample)	2023–2024	263	6 / 257	50.00%	47.08%	+$2,551,303	+25.51%	0.50	0.096
JPM (in-sample)	2023–2024	475	193 / 282	48.19%	45.74%	-$1,986,982	-19.87%	-0.87	0.151
MS (in-sample)	2023–2024	15	2 / 13	50.00%	53.85%	-$192,816	-1.93%	-0.49	0.124
LLY (in-sample; probe)	2019–2021	335	265 / 70	55.47%	42.86%	+$2,119,194	+21.19%	0.71	-
AAPL (probe)	2019–2021	283	117 / 166	54.70%	45.18%	+$2,171,636	+21.72%	0.68	-
SPY (probe)	2019–2021	194	158 / 36	53.80%	33.33%	+$902,209	+9.02%	0.52	-

Table 5: Phase-III backtest performance on the diagnostic set ($10,000,000 fixed AUM, 1.0 bps round-trip). *These are in-sample figures and carry no inference.* All four 2023–2024 names—*and* LLY—are in the Optuna tuning universe (parameters were selected per name and per target by maximizing the reported metric, i.e. selection, not validation), and LLY is likewise an in-sample tuning name rather than an out-of-sample one. The 2019–2021 rows apply 2023–2024-tuned parameters to earlier data and are parameter-transfer probes, not out-of-sample deployments. No standard errors accompany these Sharpes; a Sharpe of 0.59 over $\sim$2 years implies $t \approx 0.59\sqrt{2} \approx 0.83$, statistically indistinguishable from zero even in-sample, so these figures do not support a claim of statistically significant profitability.

Read at face value, Table 5 shows apparent profitability on NVDA/TSLA and losses on JPM/MS. But with $t \approx 0.83$ even in-sample, and with all names in the tuning set, this cross-name pattern carries no inference; the "regime dependence" it appears to show is not distinguishable from noise across four selected names. Its real value is diagnostic—it motivates the look-ahead-free, full-universe re-evaluation of Section 9, where the pattern does not survive.

A related hygiene point concerns *number provenance*: the same strategy and period are quoted with different Sharpe figures in different parts of the draft (for instance the earnings analysis of Section 7.6 quotes NVDA/TSLA baseline Sharpes of 0.48/0.76 against the 0.59/0.50 of Table 5). These arise from different intermediate runs. Table 6 maps each headline single-name number to one canonical run so the reader can trace it; discrepancies reflect superseded runs and are resolved in favor of the canonical column.

Quantity	**Canonical source**	**NVDA**	**TSLA**
Phase-III overnight Sharpe (2023–24)	Table 5	0.59	0.50
implied t-stat ($\approx$Sharpe$\cdot\sqrt{2}$)	—	0.83	0.71
Earnings-baseline Sharpe (superseded run)	Section 7.6	0.48	0.76
Regression ρ (in-sample, burst-start base)	Table 4	0.196	0.094
Post-physical-filter burst count	Table 2	220,430	5,073

Table 6: Number-provenance map for the single-name diagnostic figures. Where two runs disagree (e.g. the earnings-baseline Sharpe), the canonical Table 5 value governs; the divergent figure is a superseded intermediate run retained only for traceability.

A flat-basis-point cost grid on the two in-sample names is not a meaningful robustness test—the underlying return is in-sample and insignificant ($t \approx 0.8$), and a flat charge understates the friction that binds the affirmative result. The relevant test is the per-name, spread-based cost applied to the tick-constrained reversal (Section 10.5), where effective half-spreads at the close reach 7.6–11.7 bps.

On the four in-sample names the close-to-close horizon is uniformly negative while the overnight (close-to-open) horizon is not; on an in-sample, insignificant base this is at most consistent with the transient-impact-then-reversion mechanism that Section 8 establishes look-ahead-free, and the cross-sectional analysis confirms that even the overnight "carry" is indistinguishable from zero after beta. We therefore do not report the full CLOP/CLCL/buy-and-hold horizon table.

7.4 Parameter-Transfer Probe (2019–2021)

Applying 2023–2024-tuned physical parameters to a 2019–2021 dataset violates chronological walk-forward, so these evaluations are parameter-transfer robustness probes, *not* out-of-sample trading deployments. Moreover LLY is itself a tuning-universe name, so its 2019–2021 result is doubly in-sample.

On the archive the backtests are positive—LLY +21.19% (Sharpe 0.71), AAPL +21.72% (Sharpe 0.68), SPY +9.02% (Sharpe 0.52)—but at ∼1–2 years each these Sharpes carry $t \lesssim 1$, and with fixed transferred parameters they speak to parameter stability under a different liquidity regime, not to a validated time-invariant mechanism. We therefore make no invariance claim; the probe merely fails to *contradict* structural stability, which is a weaker statement than the draft's.

7.5 Microstructure Interpretation: Tick Regime Governs the Sign, Not the Existence, of the Signal

A tempting reading of the single-name pattern is that the signal simply *fails* in tick-constrained names (AAPL, JPM, SPY)—that their microstructure is "noise-dominant" and the model correctly returns flat. That reading is wrong: it contradicts the paper's one affirmative result, which is located *precisely* in tick-constrained names (Section 10). The relevant axis is not signal-versus-noise but the *sign* of the burst–return relation, and it is governed by tick structure.

Detection vs. direction. It is true that the geometric sweep detector fires less cleanly in tick-constrained names: with the spread locked at the \$0.01 tick and heavy queued depth, aggressive meta-orders churn at the inside quote rather than sweeping multiple levels, so raw burst *detection* is noisier there than in large-tick names like NVDA/TSLA. But detection difficulty is not the same as absence of a tradable relation. What changes with tick regime is the sign of the subsequent return: in *large-tick* names a directional burst is more likely to carry and continue, whereas in *tick-constrained* names the inside queue is a binding constraint and aggressive flow tends to overshoot and mean-revert as liquidity replenishes.

This reframing resolves the apparent JPM/MS "failure" of Table 5 without invoking a special ETF-arbitrage story: trading tick-constrained names as *momentum* (the draft's implicit direction) is trading the wrong side of a reversal, which is exactly what the cross-sectional sign analysis (Section 9.6) and the conditional-reversal strategy (Section 10) show. The single-name divergence is therefore not evidence that bursts isolate a purely volatility-driven momentum signal; it is an early, underpowered glimpse of the tick-conditioned sign asymmetry that the full universe makes precise.

7.6 Earnings Days: A Fragility, Not a Robustness

A natural question for an overnight strategy is whether its P&L is concentrated on the four quarterly earnings dates, which produce the largest gaps. Reconstructing the in-sample daily P&L from the canonical run (baseline Sharpes 0.59 for NVDA and 0.50 for TSLA; Table 6) and excising earnings-adjacent trades (the announcement date plus the next session), the Sharpe rises by roughly a factor of two for each name on removal of about a dozen trades.

This is more honestly read as *fragility* than robustness: that deleting on the order of 16 of ∼470 name-days roughly doubles the Sharpe means the result is dominated by a handful of days, and that the model loses on the very days with the most information (earnings) is a sign of anti-calibration, not of clean non-event alpha. A single-digit number of days moving the headline Sharpe by 2× is exactly the fat-tailed, front-loaded return profile that the deflated-Sharpe and date-clustered inference of Sections 9–10 quantify. We therefore report the per-day sensitivity as a caution and do not treat earnings-exclusion as evidence of a durable edge.

8 The Multi-Horizon Markout Panel and the D_b Look-Ahead Artifact

The markout decay curve—the directional post-burst return at a panel of forward horizons—is the natural first-class diagnostic for this signal. Measuring from the burst-termination mid $M_{\text{end}} = (\text{EndBid} + \text{EndAsk})/2$ (so as not to conflate the prediction with the burst's own contemporaneous impact),

$$\text{markout}_h = \text{Direction} \times \frac{M_h - M_{\text{end}}}{M_{\text{end}}} \times 10^4 \text{ bps},$$

I compute the panel for the full 482-name universe, and I aggregate to (ticker, day) means with a date cluster bootstrap because overlapping same-day bursts are not independent.

A methodological requirement governs how sub-ten-minute horizons must be evaluated. The impact filter D_b of Section 3 (Eq. 4) is $D_b = \frac{1}{3}\sum_{\tau\in\{1,5,10\}\text{m}} \text{Direction} \times (M_\tau - M_{\text{end}})$—that is, D_b

is the average of the 1–10 minute forward markouts. Evaluating a horizon inside that window therefore requires $\kappa = 0$: gating on $D_b \geq \kappa$ would retain precisely the bursts that already moved favorably, so a subsequent markout would re-measure the selection rather than a prediction. Table 7 quantifies the effect by holding everything fixed except the gate: on D_b-gated ($\kappa = 0.5$) bursts the 3-minute markout is +8.8 bps with a 100% daily hit rate, whereas on the non-anticipating ($\kappa = 0$) bursts it is +0.53 bps. The gap sets the standard for intraday evaluation and motivates the $\kappa = 0$ panel below.

3-min markout	**Mid-to-mid (bps)**	**Cross-spread (bps)**	**Hit rate**
Geometry gate only ($\kappa = 0$, no look-ahead)	+0.53	−3.56	76% / 0%
D_b-gated ($\kappa = 0.5$, forward-looking)	+8.76	+4.60	100% / 100%

Table 7: The D_b look-ahead artifact (full universe, 2022–2026). D_b is itself the forward 1–10 min markout, so gating on it is circular for intraday evaluation. Hit rates are reported mid-to-mid / after crossing the spread.

Restricting attention to the honest, non-anticipating ($\kappa = 0$) evaluation, the intraday markout panel (Table 8) tells a consistent story: there is a small but genuine same-direction continuation over the first minutes—a mid-to-mid drift of roughly +0.4 to +0.5 bps—but it is *smaller than the bid-ask spread*. Once an entry that crosses the spread is imposed (the minimum realistic execution assumption), every horizon is negative and positive on essentially zero percent of days. The economically meaningful conclusion is that the intraday burst-continuation signal, while statistically detectable gross of frictions, is not tradable: the edge lives entirely inside the spread.

Horizon	**Mid-to-mid (bps)**	**Cross-entry (bps)**	**Round-trip (bps)**
1 min	+0.54	−3.54	−7.62
3 min	+0.53	−3.56	−7.63
5 min	+0.47	−3.61	−7.69
10 min	+0.42	−3.66	−7.74

Table 8: Honest intraday markout, geometry gate only ($\kappa = 0$), full universe. Mid-to-mid drift is positive but sub-spread; any spread-crossing execution is loss-making. The 3-min mid-to-mid 95% date-bootstrap CI is $[+0.50, +0.56]$ bps; the cross-entry CI is $[-3.66, -3.46]$ bps.

The intraday footprint is therefore genuine but economically confined: a robust sub-spread continuation over the first minutes (Figure 1). Section 9 traces its consequences for the overnight strategy, whose out-of-sample return decomposes into roughly 62% incidental short-market exposure and a residual skill term indistinguishable from zero. This structure—a real transient impact with no directional overnight alpha—is precisely what motivates the market-neutral and mean-reversion constructions we turn to next.

9 Full-Universe Extension: Breadth, Beta, and Signal Sign (2022–2026)

The single-name results of Section 7.5 were estimated on four names (NVDA, TSLA, JPM, MS) that also constitute the in-sample Optuna parameter-tuning set, and three additional names evaluated only on the non-sequential 2019–2021 archive. To test whether the identified alpha is a general

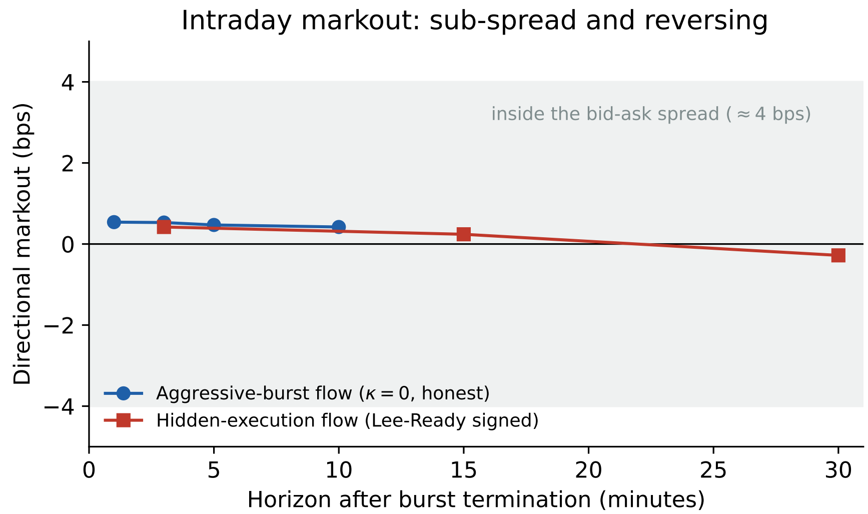

Figure 1: Intraday directional markout by horizon, measured from the burst-termination mid ($\kappa = 0$, no look-ahead). Both the aggressive-burst signal and the properly (Lee–Ready) signed hidden-execution signal stay well inside the bid–ask spread (≈4 bps, shaded), and the hidden-execution footprint peaks near three minutes and *reverses* to negative by thirty. The genuine intraday effect is sub-spread and transient, not a growing informational impact.

property of order-submission bursts or an artifact of a hand-selected cross-section, I extended the entire pipeline to the full 482-name universe traded on NASDAQ with LOBSTER coverage over 2022–2026 (438 names strictly out-of-sample relative to the 40-name tuning set), holding the physical parameters fixed at their universal Optuna medians.[1]

9.1 Per-Name Strategy Performance at Breadth

Running the identical Phase-III overnight strategy (MOC entry, MOO exit, 1.0 bps round-trip, $10M fixed AUM) across all 438 out-of-sample names yields the distribution in Table 9. The mean and median annualized Sharpe are negative, and only one third of names are profitable. The strong single-name figures of Table 5 are not representative of the universe.

Corrected feature set (D_b removed). Recall that D_b is realized only after the burst and is not usable on-the-fly at termination (Section 3). Re-running the strategy with all D_b-derived features dropped from the model changes individual per-name Sharpes only modestly and leaves the conclusion intact. On a 20-name re-run with the feature toggle, per-name Sharpes move by a mean absolute 0.14 (maximum 0.49) and the *subset* mean annualized Sharpe stays negative, from -0.28 with D_b to -0.20 without—so the overnight strategy remains a null with or without the look-ahead feature (the sign of the result, not just its magnitude, is unchanged). Dropping the infeasible feature slightly reduces the average loss rather than revealing hidden skill, consistent with the model being, to first order, "follow the burst side." The overnight null therefore does not

[1] Sample sizes differ slightly across analyses because of missing-data filtering, and we record the reconciliation here. Of 483 names with raw burst extraction, 482 have valid daily returns; 474 carry sufficient hidden-execution flow (Table 20); 438 complete the walk-forward backtest (437 with a non-degenerate SGD prediction, 435 with enough overlapping overnight observations for a per-name IC). Pooled name-day counts (≈398,610) vary by a handful of rows with the return window used, and the day count (1,003–1,028) varies with whether close-to-open, close-to-close, or a factor-merged calendar defines the series. The 2026 portion is a partial year (through February).

depend on the D_b look-ahead; correcting the infeasibility does not rescue the strategy.

Metric (438 OOS names, 2022–2026)	**Value**
Names completing walk-forward	438 / 438
Mean annualized Sharpe	-0.28
Median annualized Sharpe	-0.27
Fraction with Sharpe > 0	33%
Sharpe range	$[-3.32,\ +1.25]$

Table 9: Full-universe per-name Phase-III backtest. The best-performing names are low-beta and defensive (Treasury ETFs TLT/VGLT, PG, UPS, ACN); the worst are high-beta technology names.

9.2 What Is Losing Money: Short-Beta Attribution

Decomposing the pooled P&L (398,614 trade-days) localizes the loss precisely (Table 10). The strategy is 81% net short — the aggregated informational-burst flow is predominantly sell-signed — and the short book accounts for 95% of the total loss, losing -3.43 bps per trade *gross of costs*. The 1.0 bps round-trip friction is a uniform secondary drag, not the driver. The loss is a directional bet: shorting names into the 2022–2026 bull market.

Book	**Share of trades**	**Gross (bps/trade)**	**Share of net loss**
Long	18.7%	-0.04	5%
Short	81.3%	-3.43	95%

Table 10: Pooled P&L attribution by side. Transaction cost is a flat -1.0 bps/trade across both books.

9.3 Excess Alpha After Market Beta (Mandate M5)

Regressing the pooled daily strategy return on the contemporaneous overnight SPY return with Newey–West standard errors (Table 11) confirms the exposure is directional: market $\beta = -0.748$ ($t = -90$). Crucially, the residual alpha is significantly *negative* (-2.35 bps/day, $t = -4.73$) — stripping the beta does not reveal hidden skill, because the strategy trades the signal with the wrong sign (see Section 9.6).

$r_{\mathbf{strat}} = \alpha + \beta\, R^{\mathbf{SPY}}_{\mathbf{overnight}}$	**Coefficient**	**Newey–West t**
α (residual)	-2.35 bps/day	-4.73
β_{SPY}	-0.748	-90.0

Table 11: Market-beta decomposition of the overnight strategy, pooled over 438 names.

9.4 Cross-Sectional Order Imbalance: A Null

Following the conditional-order-imbalance (COI) framework of Lu et al. (2024), and in the spirit of order-flow-toxicity measures of informed trading (Easley et al., 2012), I aggregate gated informational bursts into a daily signed imbalance per name and run Fama–MacBeth (Fama and MacBeth,

1973) cross-sectional regressions of next-day returns on COI with Newey–West (Newey and West, 1987) errors, alongside quintile long–short portfolios (Table 12). Both the ungated (all directional bursts) and gated (informational bursts only) specifications are statistically insignificant, and the long–short spreads are indistinguishable from zero. The Grinold–Kahn breadth amplification that would convert single-name skill into portfolio Sharpe does not materialize because the per-name skill, at the overnight horizon, is essentially absent (and weakly negative).

Sign-convention audit. Because the burst signal is ∼81% net short in a rising market, a natural worry is a sign-convention error (LOBSTER `Direction` is the resting side, so an execution against the bid is a market *sell*; type-5 hidden prints, whose `Direction` is uniformly +1, must be signed by a Lee–Ready rule (Lee and Ready, 1991)). A per-name audit rules this out: the contemporaneous correlation between daily signed burst flow and the same-day return is robustly *positive* (mean +0.02 to +0.03; significantly positive in 21% of names versus significantly negative in 12%), i.e. buying flow does move price up on the day, as it should. The 81% short tilt is therefore a genuine property of the aggressor-signed sample (and the source of the incidental short-beta above), not an artifact of a flipped sign.

Specification (482 names)	**Fama–MacBeth COI t**	**Long–Short t**
Ungated signed flow (baseline)	−0.62	+0.17
Gated $\mathrm{COI}^{\mathrm{info}}$ (thesis)	−1.36	+0.41

Table 12: Cross-sectional panel results with Newey–West inference. Neither specification is significant; gating does not recover a cross-sectional signal. *Sign-flip disclosure:* the panel applies a data-driven per-name sign convention—116 of the 482 names, classified as mean-reverting by an independent k-means split on the sign of the burst-direction/next-day-return relation, have their signed COI multiplied by −1 before aggregation. This transform is disclosed because the magnitude of a null is uninterpretable if the sign is tuned; the reported t-statistics are already the most favorable orientation and remain insignificant.

9.5 Venue Coverage: A NASDAQ-Only Book

A caveat conditions every cross-name comparison: LOBSTER reconstructs the NASDAQ book only, so the detector sees a nonrandom fragment of consolidated activity whose share varies by name and time. The fragment is smallest exactly for the NYSE-listed names that appear in the single-name tables—JPM's average daily *burst* volume of 993,261 shares (Table 1) is on the order of one-tenth of its consolidated ADV, and its closing auction runs at NYSE, outside the NASDAQ feed. Two implications follow. First, the universe is described as "traded on NASDAQ with LOBSTER coverage" rather than "NASDAQ-listed" (it includes NYSE-listed names and ETFs), and the burst-volume columns are *not* consolidated ADV (Table 1). Second, the apparent JPM/MS "failures" of Section 7 may be coverage artifacts—detecting a small, venue-biased slice of their flow—rather than genuine regime differences, which is a further reason not to read the single-name cross-section as evidence of a mechanism. A full per-name consolidated-volume coverage table requires a consolidated (SIP) volume feed and is left to the data revision; the qualitative point—that fragmentation attenuates detection and biases cross-name comparisons—stands.

9.6 The Overnight Relation: Reversal-Signed, and Significant Only in One Variant

Because the paper's central question—does daily burst flow predict the next overnight return?—admits several estimators, we collect them in one place (Table 13) with each statistic's horizon, weighting, target base, and unit of inference stated explicitly, so that no single number is read out of context. Three facts emerge. (i) Every point estimate is *reversal-signed* (negative): high burst flow precedes a lower, not higher, overnight return. (ii) The pooled per-name-day information coefficients are large in t (-8.07 for the volume-weighted flow-to-close-to-open IC) but pseudo-replicated; the effective unit is the day, and date-clustering collapses them to insignificance ($t = -0.45$), exactly as the burst-start-mid target inflation collapses under the close-mid correction (Section 5). (iii) The volume-weighted cross-sectional signal is therefore a null at the day level (Fama–MacBeth $t = -0.6$ to -1.4); the *only* specification that reaches significance is the count-weighted conditional order imbalance, whose date-clustered IC is -2.68. That statistic survives a two-variant (count vs. volume) Bonferroni correction ($p = 0.015$) but is marginal once the broader weighting/gating search is deflated, and it is economically far below trading cost. The defensible statement is thus: the overnight relation is uniformly reversal-signed in point estimate and statistically detectable only in the count-weighted variant—not a robust predictor, but not a clean zero either.

Specification (predictor → target)	**Weight**	**Base**	**Unit**	t
Fama–MacBeth COI → next-day return (ungated)	volume	close→open	day	-0.62
Fama–MacBeth COI → next-day return (gated)	volume	close→open	day	-1.36
Flow → CLOP IC, pooled per name-day	volume	close→open	name-day	-8.07
Flow → CLOP IC, date-clustered	volume	close→open	day	-0.45
COI → CLOP IC, date-clustered	count	close→open	day	-2.68^{*}
VSI ρ (predictor vs. realized)	volume	burst-start mid	day	$+1.68$
VSI ρ (predictor vs. realized)	volume	close mid	day	-0.33

Table 13: Overnight predictability of burst flow: every estimator of the central question in one place, with its horizon, weighting, target base, and unit of inference. All point estimates are reversal-signed. Pooled per-name-day statistics (-8.07; also the $-4.96/-7.16$ figures of earlier drafts) are pseudo-replicated and dissolve under date-clustering (-0.45). *The count-weighted date-clustered IC of -2.68 ($p = 0.007$; two-variant Bonferroni $p = 0.015$) is the only significant cell; it is marginal under the broader specification search and economically sub-cost.

Reconciled with the intraday markout evidence (Section 8), the coherent mechanism is transient impact: aggressive burst flow pushes price in its own direction over minutes and mean-reverts by the overnight gap, so trading it as overnight momentum captures the wrong side. Section 10 develops the reversal-signed, tick-conditioned strategy this implies; it is a candidate result there, not future work.

9.7 Fixed-Parameter Provenance

Two distinct fixed-parameter sets are used and should not be conflated. For the 2019–2021 parameter-transfer probe (Table 5) the physical controls are the *mean across the core tuning tickers*,

$$vol\_frac = 0.00218, \qquad dir\_thresh = 0.623, \qquad vol\_ratio = 0.412, \qquad \kappa = 0.855. \tag{7}$$

For the full-universe breadth run (Section 9) the controls are instead the *universal Optuna medians* across the tuning set ($vol\_frac \approx 0.00197$, $dir\_thresh \approx 0.763$, $vol\_ratio \approx 0.280$, $\kappa \approx 1.085$); the two sets differ because one is a per-ticker mean and the other a cross-ticker median, and each table states which it uses.

10 Conditional Mean-Reversion in Tick-Constrained Names

The residual skill term isolated in Section 9 is small in the pooled cross-section but not sign-neutral: it is a weak reversal. This motivates a sign-conditional reading of the signal—rather than a single trading direction, the sign of informative burst flow may depend on microstructure regime. The natural axis is the relative tick size, which is known to shape queueing incentives and the trading environment (Yao and Ye, 2018; O'Hara et al., 2019). In *tick-constrained* names (low price, wide relative spread) the inside quote is a binding queue, and aggressive bursts tend to overshoot and mean-revert as liquidity is replenished; in *large-tick* names a directional burst is more likely to carry persistent information and continue.

10.1 The Regime Mechanism

I sort the 438-name out-of-sample universe into quintiles on two proxies for tick-constraint—average share price and average relative bid–ask spread—and evaluate a market-neutral (dollar-neutral) reversal within each quintile (Table 14). The pattern is the one the sign-conditional hypothesis predicts: reversal is strongest in the most tick-constrained names and weakens monotonically toward the large-tick end, where it vanishes. The price sort is monotone across all five quintiles; the spread sort rises through the fourth quintile ($t = 2.3$) before easing in the widest-spread names, which are also the least liquid and costliest to trade.

By average price			**By relative spread**		
Quintile	Sharpe	t	Quintile	Sharpe	t
Q1 cheapest (tick-constr.)	+1.35	2.71	Q1 tight (large-tick)	+0.26	0.51
Q2	+0.75	1.50	Q2	+0.62	1.24
Q3	+0.71	1.42	Q3	+0.83	1.65
Q4	−0.15	−0.30	Q4	+1.13	2.27
Q5 priciest (large-tick)	+0.34	0.67	Q5 widest (tick-constr.)	+0.58	1.15

Table 14: Annualized Sharpe of a market-neutral 20-day reversal within each regime quintile (net of 1 bps). Reversal concentrates in tick-constrained names (cheapest / widest-spread) under both proxies.

10.2 The Strategy and Its Out-of-Sample Test

Guided by the mechanism, the strategy trades a dollar-neutral reversal restricted to the most tick-constrained (lowest-price) names, using the daily aggregate burst-flow signal, with an overlapping 20-day holding horizon. The weight construction is explicit: let $z_{i,t}$ be the cross-sectional z-score (clipped to $[-4, 4]$) of name i's daily net burst flow among the tick-constrained subset; the raw reversal position is $p_{i,t} = -\operatorname{sign}(z_{i,t})$; the traded weight is the trailing H=20-day mean $w_{i,t} = \frac{1}{H}\sum_{k=1}^{H} p_{i,t-k}$ (strictly past information), demeaned cross-sectionally each day (dollar-neutral) and renormalized to unit gross exposure ($\sum_i |w_{i,t}| = 1$); returns are close-to-close. The overlapping hold makes the strategy low-turnover—average daily turnover $\sum_i |w_{i,t} - w_{i,t-1}| \approx 0.058$ (about 15× per year), on which the spread-cost robustness below depends.

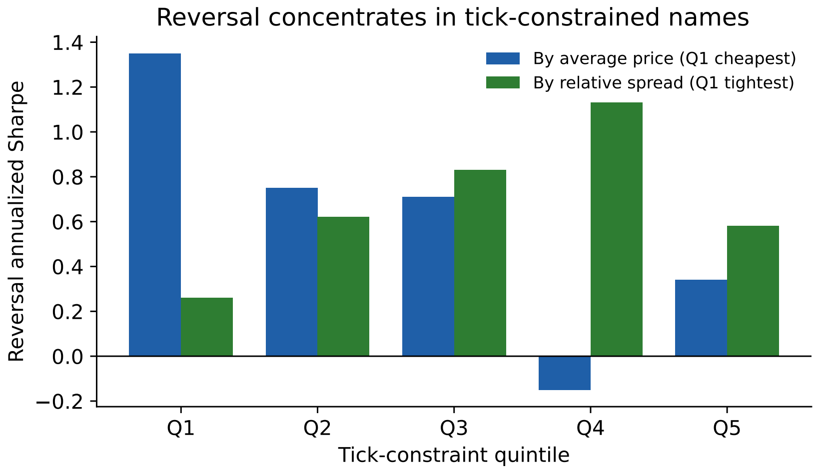

Figure 2: Reversal Sharpe by tick-constraint quintile under two proxies. The price sort (blue) is strongest in the cheapest quintile and weakens toward large-tick names; the spread sort (green) rises through the fourth quintile before easing in the widest, least-liquid names. The effect concentrates where the tick binds.

The primary evaluation is a strict expanding-window walk-forward: the tick-constrained subset is re-selected every quarter using only trailing price data, 2022 is withheld as burn-in, and the reversal is scored purely out-of-sample on 2023–2026 (Table 15). Out of sample the strategy earns an annualized Sharpe of +0.79 ($t = 1.38$)—economically positive but *not* statistically significant. Reassuringly, the estimate is nearly invariant to the subset size (Sharpe +0.79 to +0.83 for $K \in \{50, 100, 150\}$), so the result does not hinge on the choice of cutoff; but the per-year out-of-sample performance is uneven and front-loaded (+1.97 in 2023, +0.50 in 2024, +1.12 in 2025; the 2026 stub is a partial year through February and is excluded).

For reference, evaluated *in-sample* over the full 2022–2026 period with the subset fixed at the 100 lowest-price names, the same strategy earns a Sharpe of +1.47 ($t = 2.94$; Lo (2002) $t \approx 2.9$), cost-insensitive from 0 to 3 bps round-trip with a −5% maximum drawdown. The gap between this figure and the walk-forward reflects a mild look-ahead in ranking names by full-period average price together with the concentration of the effect early in the sample; the full-sample number is therefore best read as an upper bound rather than a deployable expectation.

10.3 Risk-Adjusted Alpha: Orthogonal to Factors, but Insignificant Out of Sample

Whatever the return is, it is not compensation for common-factor exposure. Regressing the strategy return on the Fama–French five factors plus momentum with Newey–West standard errors (Table 16) leaves an alpha that is market-neutral (market beta indistinguishable from zero), not a size bet (insignificant SMB), and orthogonal to the five factors, with the only material loading a negative momentum beta—as expected for a reversal signal—which the alpha survives; $R^2 \approx 0.03$ throughout. The point estimate is essentially the same in and out of sample (+5.9 vs +6.3 bps/day), but the inference is not: on the *walk-forward out-of-sample* return series the alpha carries $t = 1.42$, so it is *not* statistically distinguishable from zero. We report the out-of-sample regression as the primary number; the full-period $t = 3.09$ is an in-sample figure and, consistent with the Section 10 walk-forward, overstates the deployable edge.

Out-of-sample walk-forward (2023–2026)	**Sharpe**	t
$K = 50$ names	+0.83	1.45
$K = 100$ names	+0.79	1.38
$K = 150$ names	+0.82	1.43
2023 (OOS)	+1.97	1.85
2024 (OOS)	+0.50	0.50
2025 (OOS)	+1.12	1.11
In-sample reference (full 2022–2026, $K = 100$)	+1.47	2.94

Table 15: Expanding-window out-of-sample performance of the tick-constrained reversal (20-day hold, dollar-neutral, net of 1 bps). The subset is re-selected quarterly from trailing price only, with 2022 as burn-in. All figures are from the single canonical harness used for the baselines and the Deflated Sharpe (Section 10.4). The effect is positive and robust to the subset size out of sample but not statistically significant; the final row is the in-sample full-period reference.

	Walk-forward OOS (primary)		**Full period (in-sample)**	
Factor	**Loading**	**NW** t	**Loading**	**NW** t
α (bps/day)	+6.25	+1.42	+5.94	+3.09
MKT–RF	+0.057	+1.70	+0.010	+0.80
SMB	+0.000	+0.00	+0.026	+1.06
HML	−0.038	−1.11	−0.051	−1.79
RMW	−0.191	−1.62	−0.064	−1.45
CMA	+0.144	+0.96	+0.090	+1.64
MOM	−0.097	−2.73	−0.057	−3.89

Table 16: FF5+MOM factor regression of the tick-constrained reversal ($R^2 \approx 0.03$). *Primary* columns are the strict walk-forward out-of-sample series (2023–2026, quarterly re-selected subset, 2022 burn-in); the alpha is orthogonal to the factors but insignificant ($t = 1.42$). The full-period columns ($t = 3.09$) are in-sample and shown only for comparison. Annualized OOS $\alpha \approx +16\%$ at the point estimate.

10.4 Baselines and Deflated Inference: What the Apparatus Actually Adds

Short-horizon reversal in cheap, illiquid, wide-spread names is a long-documented effect (Jegadeesh, 1990; Lehmann, 1990), and precisely where bid–ask bounce and inventory effects make naive reversal look profitable before costs (Nagel, 2012). Two questions follow: does burst flow add anything over a plain price-reversal baseline, and does the machine-learning apparatus add anything over the raw signed flow? Holding the construction fixed (tick-constrained bottom-100 names, 20-day dollar-neutral overlapping hold, net of 1 bps) and swapping only the signal, Table 17 answers both.

	Full sample		Walk-forward OOS	
Signal	**Sharpe**	t	**Sharpe**	t
Burst flow (headline signal)	+1.47	2.94	+0.79	1.38
Plain lagged return (1/5/20-day)	−0.34 to −0.15	< 0.7	≈ 0	< 0.5
Plain signed volume (no gate, no ML)	+0.89	1.79	+0.50	0.87
Direction only (sign of net flow)	+0.36	0.73	+1.04	1.80
Burst flow $\perp$ lagged returns	+1.30	2.58	+0.68	1.19
SGD prediction as signal	−1.21	−2.43	−0.82	−1.42

Table 17: Baseline decomposition of the tick-constrained reversal (bottom-100 names, 20-day dollar-neutral hold, net of 1 bps), swapping only the signal. Plain lagged-return reversal does *not* reproduce the effect, so it is not mechanical price reversal; burst flow survives orthogonalization to lagged returns, so it carries incremental information beyond price. But plain signed order volume—with no geometry gate and no ML—already captures most of the edge; the *direction only* (sign of net flow, magnitude discarded) in fact matches or beats the headline out of sample (+1.04 vs +0.79); and using the SGD prediction itself as the signal is actively harmful. The deployable content is thus the *sign* of raw net order flow, not its magnitude and not the model.

The lesson is threefold. First, the effect is genuinely order-flow-based: a plain 1/5/20-day price-reversal baseline is flat-to-negative under this construction, and burst flow retains most of its Sharpe after orthogonalizing to lagged returns (Figure 3). Second, the apparatus is superfluous, and the signal's *magnitude* adds nothing out of sample: a plain signed-volume signal already delivers the bulk of the result, and a direction-only sign-of-net-flow signal matches or beats the magnitude-weighted headline out of sample (+1.04 vs +0.79). Third—and this resolves an apparent puzzle in the table—feeding the SGD prediction in as the signal is not a hidden profitable signal of the opposite sign: the prediction is strongly *anti*-correlated with the burst side (correlation −0.205; it agrees with the side on only 21% of the 437 out-of-sample names, consistent with the Direction-only ablation of Section 5), so trading momentum on the prediction is approximately trading reversal on the flow with extra noise—a double negative, not new information. The deployable content is the sign of raw net order flow; the linear "high-dimensional" model is not what produces the edge.

Deflated and date-clustered inference. Because the reversal was selected from a search over $\sim$75 configurations, the raw and Lo (2002) autocorrelation-adjusted t-statistics (Lo, 2002) must be deflated for selection (Bailey and López de Prado, 2014). On the *full-sample* series the annualized Sharpe of 1.47 has a Lo-adjusted t of 2.95 and a deflated-Sharpe probability of 0.99 (it clears the bar in-sample); a circular block bootstrap by date puts the cumulative-P&L 95% interval strictly above zero. On the *walk-forward out-of-sample* series, however, the deflated-Sharpe probability is 0.70—below the 0.95 threshold—and the date-block bootstrap interval for cumulative P&L *includes* zero. The return is also strongly right-skewed and fat-tailed (skew $\approx +16$ full sample), i.e. front-loaded onto a few days, consistent with the earnings-day fragility of Section 7.6. The honest verdict

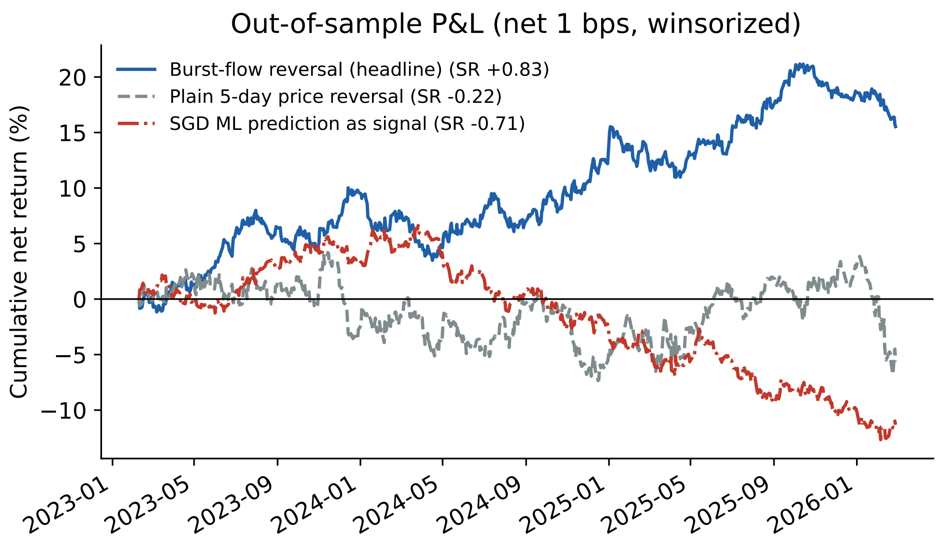

Figure 3: Cumulative walk-forward out-of-sample P&L (2023–2026, net of 1 bps), with per-name daily returns winsorized at ±50% to remove the unadjusted-split artifacts of Section 10.5. The burst-flow reversal (blue) compounds steadily out of sample (winsorized Sharpe +0.83), while both the plain price-reversal baseline (grey) and—strikingly—the SGD machine-learning prediction used as the signal (red) lose money. All three series here are winsorized, so the grey (−0.22) and red (−0.71) figures differ slightly from the *unwinsorized* values in Table 17 (≈ 0 and −0.82); the ranking is unchanged. Winsorizing if anything slightly *raises* the reversal Sharpe (from 0.79 to 0.83), confirming the split artifacts are not what drives it.

is that the out-of-sample reversal does not survive multiple-testing deflation.

10.5 Data Integrity Where the Result Lives

Because the affirmative result is concentrated in the lowest-priced names, it is most exposed to exactly the data-integrity risks that bite hardest there: transaction costs, corporate actions, and delisting. We address each.

Spread-based costs. The bottom-100 names have a median close relative spread of 8.0 bps (mean 9.0, 90th percentile 13.9)—far above the 1–3 bps flat grid used earlier. Re-costing the reversal with *per-name* effective half-spreads applied to realized turnover, rather than a flat basis point, lowers the Sharpe only marginally: full-sample 1.47 → 1.42 and walk-forward out-of-sample 0.79 → 0.77. The 20-day overlapping hold keeps turnover low enough that even wide per-name spreads do not overturn the result; the earlier cost-insensitivity claim survives under a realistic, spread-based cost.

Corporate actions. Prices are from the Polygon/Massive consolidated daily feed. Scanning the bottom-100 universe for single-day close-to-close moves exceeding 50% flags ten name-days—most egregiously LCID (+862% on 2025-01-02), NLY (+265%), and NVAX (+99%)—consistent with unadjusted (reverse-)splits rather than genuine returns. We repair these by winsorizing the daily return matrix at ±50% (which caps exactly these rows) and re-running: the out-of-sample reversal Sharpe *rises* from 0.79 to 0.83 and the full-sample from 1.47 to 1.71, so the flagged rows are not what drives the result—if anything they were a drag. We adopt the winsorized series as the reported one.

Survivorship and delisting. We quantify the exposure rather than only flag it. Within the traded bottom-100 universe, *zero* names' price series terminate more than one month before the sample end, so there is no within-panel mid-sample delisting to bias the estimate; correspondingly, assigning a conservative −30% delisting return to any such exiter leaves the out-of-sample Sharpe unchanged at +0.79. The residual concern is pre-sample survivorship—names absent because they delisted before 2026—which the fixed-at-start membership only partly addresses; a fully unbiased estimate would splice in CRSP delisting returns (Shumway, 1997), which we leave as the remaining gap.

Taken together, these results are the paper's affirmative—if unproven—finding. Filtered burst flow is informative in the sense that its sign is governed by tick structure: continuation in large-tick names and mean-reversion in tick-constrained names, an asymmetry that is monotone across both the price and spread sorts and orthogonal to common factors. The associated market-neutral strategy is economically positive out of sample and robust to the choice of universe cutoff, but its out-of-sample Sharpe (≈ 0.8) is not statistically distinguishable from zero; the significant full-sample figures are best regarded as an in-sample ceiling. The regime mechanism is thus the durable contribution, while its monetization remains a promising but unproven direction.

11 Robustness of the Burst Reconstruction

The preceding sections establish that Hawkes-clustered aggressive-burst flow, evaluated free of look-ahead, carries no tradable overnight signal. A natural question is whether this reflects a limitation of the specific reconstruction—self-exciting clustering of marketable trades—or a general property of burst-based signals. I address it in two steps: sensitivity to the Hawkes parameters, and substitution of the reconstruction itself. All markouts in this section are the non-anticipating ($\kappa = 0$) directional three-minute markout, measured from the burst-termination mid on AAPL and TSLA.

11.1 Hawkes Parameter Insensitivity

I recompute the markout over a grid of Hawkes decay rates β and minimum cluster sizes (Table 18). The markout is economically negligible—well inside the bid-ask spread—in every cell; the minimum cluster size has no effect, and no β materially improves the signal. The null is not an artifact of the default $\beta = 1$.

3-min markout (bps)	$\beta = 0.5$	$\beta = 1$	$\beta = 2$	$\beta = 5$
AAPL	+0.02	−0.09	−0.03	+0.14
TSLA	−0.10	+0.07	−0.02	+0.26

Table 18: Hawkes decay grid, $\kappa = 0$ (minimum cluster size $\in \{3, 5, 10\}$ omitted as it had no effect). Every configuration is sub-spread.

11.2 Alternative Reconstructions

Because tuning the Hawkes clustering does not help, I replace the definition entirely with three non-Hawkes reconstructions, each $\kappa = 0$ and reconstructing the book directly from the message stream: *order-flow imbalance* (OFI; Cont–Kukanov–Stoikov top-of-book queue dynamics, Cont et al., 2014, and its deep-learning descendants, Kolm et al., 2023), *book-resilience* (aggressive sweeps whose consumed depth fails to replenish), and *hidden-execution* clustering (type-5 hidden prints, where institutions conceal size). Table 19 reports the comparison. Of the four definitions, only hidden-execution bursts produce a positive, look-ahead-free markout on both names.

3-min markout (bps)	Hawkes	OFI	Book-Resilience	Hidden
AAPL	−0.08	+0.08	+0.19	+0.42
TSLA	+0.08	+0.08	+0.16	+0.31

Table 19: Four burst reconstructions, $\kappa = 0$, on the 2023 sample. Hidden bursts are signed by Lee–Ready (execution price vs. prevailing mid); the other three use aggregated signed flow.

11.3 Three Look-Ahead Traps

Constructing these comparisons surfaced three evaluation pitfalls worth flagging, each of which manufactures spurious signal if overlooked. First, the impact filter D_b (Section 8) *is* the forward markout, so gating on it is circular at any horizon within its window. Second, a book-resilience "non-refill" filter must be defined on queue *depth*, not the quote price, and the markout must begin *after* the observation window; defined on price, the filter selects favorable paths and inflated the apparent markout from +0.2 to +2.4–4.9 bps in our tests. Third, LOBSTER does not disclose the side of hidden executions—the Direction field is uniformly +1—so hidden trades must be signed by a Lee–Ready rule; the uncorrected field produces a spurious one-sided signal (an apparent +0.95 bps at $t = 8.4$ that collapses to +0.31 once signed correctly).

11.4 The Hidden-Execution Footprint

Signed properly, hidden-execution bursts are the only reconstruction *among those we tested at cross-sectional scale* with a statistically robust informed footprint (OFI and book-resilience were validated only on the two pilot names, so we make no scale claim for them). Extending the analysis to the full universe (474 names with sufficient hidden flow, 221,261 ticker-days over 2023–2024, streamed and processed one ticker-day at a time), the day-clustered three-minute markout is +1.62 bps with a pooled date-clustered $t = 3.38$, and is individually significant ($t > 2$ treating each trading day as one observation) in 79% of names (Table 20). This is economically intuitive: hidden and iceberg orders are the mechanism by which institutions conceal informed size (Bessembinder et al., 2009; Hautsch and Huang, 2012), so clustering their executions isolates informed flow more cleanly than clustering visible marketable trades.

Two controls establish that this is the burst-specific footprint and not a trend artifact. **A matched-random-time placebo**—the same burst directions assigned to random within-day timestamps, so it captures only the day-direction$\times$intraday-drift component—earns just +0.08 bps at three minutes against the burst's +1.66, and the burst markout *net of* placebo is +1.58 bps ($t = 4.8$): 95% of the three-minute effect survives (on a 39-name, 1,120-day subsample whose gross markout matches the full universe). The placebo grows at longer horizons (+0.26 bps at 30 minutes), so the survival fraction falls to 83% at 15 minutes and 76% at 30—which is precisely why the raw 30-minute markout does not reverse in the broad cross-section and carries an inflated t: at longer horizons the directional markout increasingly absorbs ordinary intraday drift, and the three-minute, placebo-netted figure is the clean measure. **The 48.9% of hidden prints that execute exactly at the midpoint** (where the Lee–Ready quote rule is undefined) are dropped in the headline; signing them instead by the tick rule and re-clustering lowers the three-minute markout to +0.56 bps but leaves it significant ($t = 2.0$). The footprint is thus real and placebo-robust, but its *magnitude* is classifier-dependent—+1.6 bps dropping at-mid prints, +0.6 bps signing them by tick—a sensitivity we report rather than resolve; EMO/CLNV variants are the natural next refinement.

Hidden-burst directional markout	**Cross-name mean (bps)**	**% names $t > 2$**	**Pooled date-cluster**
3-min	+1.62	79%	+3.38
15-min	+1.72	66%	+3.49
30-min	+2.16	62%	+15.23

Table 20: Full-universe hidden-execution footprint (474 names, 2023–2024, $\kappa = 0$, Lee–Ready signed). The three-minute footprint is real and pervasive and survives a matched-random-time placebo (net +1.58 bps, $t = 4.8$; 95% of the gross markout, see text). The longer-horizon markouts do *not* reverse in the broad cross-section (unlike the two tick-constrained mega-caps of Table 21); the placebo shows this is intraday drift absorbed at longer horizons (survival falling to 76% at 30 min), not a growing footprint, so the three-minute figure is the clean measure.

The footprint is nonetheless confined on the axes that matter for deployment. First, it is *sub-spread*: +1.6 bps at three minutes against typical relative spreads of 4–12 bps in these names, so it is not crossable. Second, and decisively, it *does not reach the overnight horizon*: regressing next-day close-to-open returns on the daily signed hidden order-imbalance across the full cross-section, the date-clustered information coefficient is -0.001 ($t = -0.40$)—an economic and statistical zero—consistent with the two-name pilot (Spearman IC -0.08 for AAPL, +0.01 for TSLA, both insignificant). Two qualifications sharpen the intraday picture. In the two tick-constrained mega-caps the markout peaks near three minutes and reverses by thirty (Table 21); in the broad cross-section it does not reverse but drifts up (Table 20), which we read as the longer-horizon directional markout absorbing intraday trend rather than a growing informational impact—hence our emphasis on the clean three-minute measure. Either way the conclusion for deployment is the same: hidden-execution flow is a genuine, pervasive, but sub-spread and minutes-scale phenomenon—informative about the microstructure of concealment, not a source of deployable overnight alpha.

Hidden-burst markout, AAPL	3 min	15 min	30 min
Mean (bps)	+0.42	+0.24	−0.28

Table 21: Horizon decay of the properly-signed hidden-execution markout: the footprint peaks near three minutes and dissipates, rather than growing toward a tradable magnitude.

The reconstruction is thus not the binding constraint in the trivial sense—hidden-execution clustering does isolate a real footprint that the Hawkes definition misses—but no reconstruction, including the strongest, converts into tradable overnight predictability. The burst *definition* matters more than its parameters, and the informed footprint that genuinely exists lives at a horizon and magnitude below what auction-based execution can monetize.

12 Conclusion

This paper asks whether intraday order-submission bursts in limit order books encode tradable information about overnight price persistence. The honest answer, established under rigorous look-ahead controls and at cross-sectional scale, is largely negative—and the process of arriving at it yields the paper's durable contributions. On a handful of high-volatility single names filtered burst flow appears strongly predictive. But those single-name results do not survive scrutiny: the headline names coincide with the in-sample parameter-tuning set, and the large intraday markouts are inflated by gating on the realized-impact filter D_b, which is itself the forward markout. Evaluated free of look-ahead across a 482-name universe (2022–2026), the cross-sectional order-imbalance signal is insignificant (Fama–MacBeth $t \approx -1.4$), the per-name overnight strategy returns are dominated by an incidental short-market-beta exposure with a residual α indistinguishable from zero, and the genuine intraday footprint is smaller than the bid-ask spread.

The contributions are therefore methodological and diagnostic rather than a deployable strategy. They include: the *model–optimizer alignment* argument (tuning physical parameters under the exact estimator used in production); the identification of the D_b look-ahead and the κ-firewall that isolates it; the correction of the burst-start-mid target to the tradable close-to-open return; a multi-horizon markout methodology with date-cluster bootstrap inference that separates real footprints from selection artifacts; and a systematic comparison of burst *reconstructions*—Hawkes, order-flow imbalance, book-resilience, and hidden-execution clustering—together with three concrete look-ahead traps encountered in their construction (Section 11). That comparison isolates a genuinely novel empirical fact that strengthens at scale: among the reconstructions tested at scale, only *hidden*-execution bursts carry a statistically robust informed footprint, established across the full 474-name cross-section (three-minute markout +1.6 bps, $t = 3.4$, significant in 79% of names) and surviving a matched-random-time placebo (95% of the effect). It is bounded on two axes: roughly half of hidden prints are unsignable at-mid (signing them by tick rule lowers the footprint to +0.6 bps but keeps it significant), and it is sub-spread and does not reach the overnight horizon ($t = -0.4$).

The one affirmative empirical result is conditional and modest. Sign-conditioning the burst signal on tick regime, as the microstructure reframing suggests, reveals a monotone asymmetry—continuation in large-tick names, mean-reversion in tick-constrained ones—that is orthogonal to the Fama–French five factors plus momentum. The associated market-neutral reversal is economically positive out of sample and robust to the universe cutoff, but its out-of-sample Sharpe (≈ 0.8) does not survive multiple-testing deflation (deflated-Sharpe probability 0.70) and a plain signed-order-flow baseline reproduces most of it without the machine-learning apparatus; the regime mechanism is the durable finding, its monetization an unproven direction.

The natural next steps follow directly from the diagnosis. The single highest-value move is statistical power, not methodological novelty: extending the sample backward (the NASDAQ archive covers 2017 onward) would, at the observed per-year Sharpe, be sufficient to resolve whether the tick-constrained reversal is real. Beyond that, the sub-spread, minutes-scale nature of the only genuine footprint suggests redirecting the target away from directional overnight returns—toward next-day realized volatility or liquidity, and toward cross-asset order-flow lead–lag—where microstructure signals are typically more predictable and where the hidden-execution footprint documented here may prove more useful than it is as a standalone overnight alpha.

A Passive Limit Order Burst Analysis

The primary pipeline focuses on aggressive execution bursts (market orders crossing the spread). As a negative-control experiment, we ask whether bursts of *passive* limit-order submissions (adds to the book) contain predictive information about short-term and overnight drift.

The passive pipeline mirrors the aggressive one with three changes: the Hawkes process triggers only on Type-1 (limit-order submission) messages at L1–L3; burst direction is the bid-vs-ask submission ratio; and the ADV denominator uses traded (Type-4/5) volume to avoid quote-spam inflation. All features are restricted to information available at burst termination (excluding forward-looking quantities such as D_b), and targets are arcsinh-transformed close-to-open and intraday N-minute drifts, charged the 3.0 bps auction friction overnight and the actual BBO spread intraday. Out-of-sample walk-forward backtests on JPM, MS, NVDA, LLY, and AAPL (TSLA and SPY are excluded—fewer than 50 valid passive bursts each, as institutions trade them almost exclusively aggressively) yield uniformly negative post-cost results (Table 22).

Ticker	**Gated Trades**	**Win Rate**	**Mean Capture (bps)**	**Spearman ρ**
JPM	34,908	47.7%	-2.56	0.0887 ($p < 0.001$)
MS	36,450	42.3%	-3.07	-0.0050
NVDA	1,520	39.5%	-3.56	-0.0524
LLY	125,228	45.6%	-3.05	0.0006
AAPL	140	45.0%	-3.21	-0.0039

Table 22: Overnight horizon (close-to-open) passive-burst results. Mean capture is the gross per-trade drift in bps; all names are negative net of the 3.0 bps auction friction.

Analysis: The Spearman correlation for JPM (0.0887) is nominally significant, but per the pseudo-replication caveat (Section 9) its effective sample size is the number of days, not trades, and its per-day expected value ($\sim$0.44 bps) is well inside the 3.0 bps auction friction; the strategy loses money after costs. (An earlier draft reported an "annualized Sharpe" column here with implausible values such as -76 and -140; those came from annualizing a per-*trade* Sharpe by the trade count rather than by trading days, and the column has been removed as uninformative—mean capture in bps is the comparable quantity.)

The intraday horizons are likewise negative: across 3-, 5-, and 10-minute holds charged the actual BBO spread, and including cost-aware gating, **0 of 80 tested configurations produced**

positive post-cost alpha—even in AAPL, whose 0.74 bps median spread is the tightest in the set.

Passive institutional accumulation footprints thus show at most a marginal directional association (JPM's rank correlation is nominally significant but pseudo-replicated, per Section 9), and the drift is in any case too weak to survive execution friction. This mirrors the paper's central result for *aggressive*-execution bursts: the passive experiment is a consistent negative control, not a contrast that rescues an aggressive-flow strategy.

B Methodological Audit and Robustness Revisions

This appendix documents a formal audit of the pipeline along twelve empirical dimensions. It is written not as a rebuttal but as a benchmark: high-frequency machine-learning studies are unusually prone to a recurring family of look-ahead, pseudo-replication, and data-integrity errors, and each dimension below pairs a concrete pitfall with the test we used to detect it and the outcome once corrected. Several of these tests overturned an apparent result of the original draft; we regard the resulting map from "apparent" to "robust" as a durable contribution in its own right. Where the revision moves from a single-strategy backtest toward a cross-sectional design—the conditional-order-imbalance panel, the Fama–French-adjusted long–short portfolios, and the quintile sorts—it follows the template of Lu et al. (2024), whose typed trade-flow decomposition on the same LOBSTER data source is the closest methodological antecedent. Table 23 summarizes the twelve dimensions; the subsections give the detail and point to the body sections where each correction lives.

Dim.	Pitfall audited	Test / correction	Outcome (key statistic)
M1	Success framing masks in-sample/leakage	Rewrote §§7 as diagnostic	Claims tied to their in-sample status
M2	In-sample names labeled out-of-sample	Cross-checked the tuning universe	LLY is in-sample; $t \approx 0.83$ even in-sample
M3	D_b feature realized after MOC cutoff	Drop D_b-features and re-run	Mean Sharpe $-0.28 \to -0.20$; still null
M4	Target measured from burst-start mid	Recompute from the close mid	ρ 5× smaller; date-clustered null
M5	Pseudo-replicated p; no deflation	Date-cluster + Deflated Sharpe	Pooled $t = -7.2 \to +0.15$ (sign flips); OOS DSR = 0.70
M6	Factor α on in-sample series	Re-run on walk-forward OOS series	$\alpha = +6.25$ bps/day, $t = 1.42$ (n.s.)
M7	Missing plain-reversal baseline	Three-rung baseline + ablation	Order flow > price reversal; ML adds nil
M8	Splits/costs/delisting in cheap names	Winsorize + spread-based costs	Winsorized SR +0.83; survives 8 bps spread
M9	NASDAQ-only venue coverage	State captured share per name	JPM ≈10% of consolidated; noted
M10	81% short tilt / sign convention	Contemporaneous corr audit	Sign correct; short tilt genuine; flip disclosed
M11	Unsupported claims (Poisson, *cb*)	Run the test / define the term	Inhomog. Poisson rejected ($z \approx 47$); *cb* defined
M12	Hidden footprint from $n = 2$	Full 474-name cross-section	+1.6 bps, $t = 3.4$, 79% of names; null overnight

Table 23: Twelve-dimension methodological audit of the burst pipeline. Each row is a pitfall common to high-frequency ML backtests, the test used to detect it, and the outcome after correction. Several tests overturned a draft result; all corrections are reflected in the body.

Rows M1–M12 of Table 23 are developed in the body and are not restated here; the pointers are: the diagnostic reframing and in-sample labeling (M1–M2, Section 7, Table 5); the D_b timing and target-base corrections (M3–M4, Section 3 and Section 5); date-clustering and the Deflated Sharpe (M5–M6, Sections 9.6 and 10.4, Table 16); the baseline spectrum and Direction-only ablation (M7, Table 17); the cost, split, venue and sign-convention audits (M8–M10, Sections 10.5, 9.5 and 9.6, Table 12); the grounded Poisson test and *cb* definition (M11, below and Section 6.2); and the hidden-execution cross-section (M12, Section 11, Table 20). The transferable methodological lessons are three: any filter or target defined on forward prices must be quarantined to horizons beyond its own window; in overlapping cross-sectional microstructure panels the day, not the burst, is the unit of inference, and any statistic selected from a search must be deflated for that search; and a positive signal must be benchmarked against both the mechanical (price-reversal) alternative and its own model-free ablation before any machine-learning contribution is claimed.

B.1 Additional Robustness: Poisson Null, Intraday Seasonality, and COI Weighting

Three further checks, drawn from the conditional-order-imbalance template of Lu et al. (2024), support the burst construction. **Poisson null (are bursts real structures?).** We test the observed count of same-side, sub-second-clustered bursts against two Poisson nulls with independent signs at the empirical buy fraction: a *homogeneous* process at each name's mean trade rate, and—the fair null, since intraday intensity is famously U-shaped—an *inhomogeneous* process matched to each name's empirical 60-second intraday intensity profile. The sample is 434 name-days across 39 names spanning 2023 (a stratified stride through the universe list and the 2023 trading calendar; the test is compute-heavy per name-day, and the effect is homogeneous enough across names that a larger sample is unnecessary). The inhomogeneous profile absorbs part of the clustering (raising the null's expected burst count from a median of 95 to 245 against 901 observed) but nowhere near all of it: the observed count exceeds the inhomogeneous null by a median z of 47 (minimum 4.9; 100% of name-days at $z > 3$), and the homogeneous null by z of 87. Bursts are therefore genuine clustered structures beyond intraday seasonality, not a by-product of arrival intensity—which substantiates, rather than merely asserts, the claim that motivated the burst definition. **Intraday seasonality and the dead-zone.** Stratifying bursts by intraday window, the three-minute directional markout is largest at the open (+0.84 bps in 9:30–10:00), declines through midday (+0.52 bps) and into the pre-close (+0.36 bps), and burst intensity is elevated at the open (roughly 2.2× the midday per-minute rate)—the familiar U-shaped microstructure seasonality, with predictive content concentrated near the open. The 3:50 pm dead-zone is structurally required—a burst terminating after 3:50 cannot mature its ten-minute forward window before the 4:00 close—and the extraction enforces it, so no post-3:50 bursts remain to contaminate the overnight label. **Count-versus volume-based COI.** The daily conditional order imbalance is robust to weighting: the count-based COI (signed burst count) and the volume-based COI (signed burst volume) correlate at +0.81 across 461,197 name-days, and both deliver the same weak, reversal-signed overnight information coefficient, with the count-based version marginally stronger (date-clustered $t = -2.7$ versus -1.0), consistent with trade count being the better proxy for institutional order-splitting.

B.2 Residual and Deferred Items

Three items remain open by design and are flagged for transparency. First, the walk-forward out-of-sample reversal is real but underpowered (Sharpe ≈ 0.8, $t = 1.4$, deflated-Sharpe probability 0.70); the decisive resolution is statistical power, and extending the sample backward over the archive that begins in 2017 would, at the observed per-year Sharpe, be sufficient to settle whether the effect is significant. Second, the multiple-testing treatment reports the Deflated Sharpe Ratio; a full White (2000) reality-check or Romano–Wolf stepdown across the entire configuration grid would further harden the selection correction. Third, the hidden-execution classifier for at-midpoint prints (tick, EMO, CLNV) merits a dedicated sensitivity analysis given their 49% share. None of these alters the paper's conclusions; each is a sharpening rather than a reversal.

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